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TRENCH WALL DEVICE AND METHOD FOR CREATING A TRENCH IN THE GROUND

Abstract

The invention relates to a diaphragm wall device for creating a trench in the ground, comprising a base frame which extends in a longitudinal direction, at least one soil removal device which is arranged on a lower end region of the base frame, a holding device for holding the diaphragm wall device on an upper end region, a bracing device with at least two opposite bracing elements displaceable laterally, transversely to the longitudinal direction, and at least one actuating element, by way of which the tensioning elements can be extended laterally in the trench for bracing at least a part of the diaphragm wall device, an intermediate frame which is mounted displaceably along the base frame at a lower area of the base frame and on which the bracing device with the at least two bracing elements and the at least one actuating element is arranged, and a pressing-on device, by means of which a at least one soil removal device can be applied with a pressure force, wherein the pressing-on device is arranged between the intermediate frame and the base frame. According to the invention, the holding device is fixedly connected with the intermediate frame and the base frame with the soil removal device is displaceable with respect to the holding device.

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Background/Summary

[0001] The invention relates to a trench wall device for creating a trench in the ground, comprising a base frame which extends in a longitudinal direction, at least one soil removal device, which is arranged on a lower end region of the base frame, a holding device on an upper end region of the trench wall device, a bracing system with at least two opposite bracing elements displaceable laterally, transversely to the longitudinal direction and with at least one actuating element, by means of which the bracing elements can be extended laterally in the trench for bracing at least a part of the trench wall device, an intermediate frame, which is displaceably mounted along the base frame in a lower region of the base frame and on which the bracing system with the at least two bracing elements and the at least one actuating element is arranged, and a pressing-against device, by means of which the at least one soil removal device can be applied with a contact pressure force, wherein the pressing-against device is arranged between the intermediate frame and the base frame, according to the preamble of claim 1.

[0002] The invention also relates to a method for creating a trench in the ground comprising such a trench wall device according to the preamble of claim 13.

[0003] Trench wall devices of this kind are in particular known as so-called trench cutters comprising cutting wheels as soil removal devices. These are usually suspended on a carrier device by means of a support cable. A holding device for a support cable, a support bar or the like is arranged on the upper end of the base frame. As a result, with such devices, the maximum contact pressure force on the ground is limited by the weight of the trench wall device.

[0004] When working on harder ground, for example stones or rocks, it may, however, be necessary to apply a greater contact pressure force to the cutting wheels. In principle, this can be carried out from outside of the cut trench by means of lifting tables or corresponding hydraulic cylinders on the carrier device. In this case, however, a corresponding solid rod assembly is required for transmitting the axial forces to the trench wall device, what is overall very expensive and is no longer feasible at greater cutting depths.

[0005] EP 3 347 526 A1 discloses a trench cutter in which longitudinally and transversely displaceable bracing plates, which form a unit together with transversely directed bracing cylinders, are provided on a cutter frame. Here, the bracing cylinders are each supported on the opposite bracing plates. The unit formed by bracing plates and bracing cylinders is mounted, so as to be movable relative to a base frame of the cutter in the transverse direction, such that floating mounting of the unit as a whole is provided. When bracing the cutter in a cut trench in the transverse direction by extending the bracing cylinders, this floating mounting prevents the base frame suspended on a support cable from shifting its position. To increase a contact pressure force on the cutting wheels, an arrangement of vertically directed pressure cylinders is provided below the unit comprising the bracing plates.

[0006] This known arrangement requires a relatively large installation space, wherein the bracing plates are arranged at a considerable spacing above the cutting wheels in a central region of the frame. When creating a cut trench in the ground, an additional contact pressure force can thus only

be applied at a relatively late stage, when a trench of several metres in depth has already been created.

[0007] EP 0 593 355 A1 teaches a similar floating mounting of bracing plates in a transverse direction. The bracing plates are arranged on a shared piston and are therefore displaceable together in the transverse direction.

[0008] A trench cutter comprising a base frame and a lower, vertically displaceably mounted tool carrier part comprising the cutting wheels is known from EP 0 811 724 B1. The base frame held on a support cable can be braced in the cut trench by means of transversely directed bracing elements while the cutting wheels are movable downwards.

[0009] EP 3 401 444 B1 discloses a trench cutter in which transversely directed actuating elements are likewise supported both on a first bracing plate and on a second, opposite bracing plate. In addition, obliquely downwardly directed additional bracing cylinders are provided, by means of which the individual bracing plates can be additionally braced relative to an intermediate frame. The intermediate frame is displaceably guided on a base frame of the cutter in propulsion direction. By means of upper, obliquely directed contact pressing cylinders, a tensile force can be applied to the cutting wheels, in order to apply an additional propulsive force. In this arrangement too, a holding device for a support cable is attached to the base frame.

[0010] DE 38 05 868 discloses a slurry wall device according to the preamble of claim 1. This slurry wall device comprises a relatively large and solid “intermediate frame” on the upper end of a “base frame”. This “intermediate frame”, with its large mass, cannot contribute any (weight) force whatsoever in the propulsion direction in its braced state. Therefore, this mass is ineffectual and ineffective in the braced state.

[0011] The problem addressed by the invention is to specify a trench wall device and a method by means of which a trench in the ground can be created efficiently, with the slurry wall device having a compact structure.

[0012] The problem is solved both by a trench wall device having the features of claim 1 and by a method having the features of claim 13. Preferred embodiments of the invention are set out in the dependent claims.

[0013] The trench wall device according to the invention is characterised in that the holding device is fixedly connected to the intermediate frame and that the base frame with the soil removal device is displaceable relative to the holding device.

[0014] A first aspect of the invention resides in that the holding device for holding the trench wall device is not directly attached to the base frame with the soil removal device, but instead is to be attached to the intermediate frame with the bracing element. The base frame with the soil removal device is thus not only displaceable relative to the intermediate frame with the bracing element, but also relative to the holding device in propulsion direction. Therefore, a propulsion of the soil removal device, in particular of cutting wheels, on the base frame is brought about without the holding device needing to be repositioned. The holding device is provided for connecting, in particular to a support cable or support bar on a carrier device. It is therefore possible to displace the soil removal device without a support cable or the support bar needing to be repositioned on the carrier device. This simplifies the control of the feed motion and a desired contact pressure force. This can then take place, in particular by actuating only the contact pressing device.

[0015] According to another aspect of the invention, the intermediate frame with the bracing system is arranged in a lower region of the trench wall device. Therefore, even with a relatively low amount of excavation in advance, the trench wall device can be braced to the walls of a previously formed guide trench. After the bracing by the bracing system, the base frame is guided on the intermediate frame in a substantially vertical direction and a desired contact pressure force can be applied to the soil removal device even at an early point in time.

[0016] According to a further development of the invention, it is preferred that the holding device is fixedly connected to the intermediate frame via one or multiple longitudinal beams, which

extend along the base frame. In principle, just one single longitudinal beam can be sufficient, via which the intermediate frame arranged in a lower region is fixedly connected to the holding device on the upper region of the trench wall device. For force distribution that is as symmetrical as possible, preferably two or multiple, in particular four, longitudinal beams are preferably arranged along outer sides of the base frame. The longitudinal beams can each, in particular be constructed as double-T beams. Preferably, corresponding trough-shaped recesses are provided along the base frame for the longitudinal beams, such that they do not protrude beyond the outer dimensions of the base frame.

[0017] A particularly robust configuration of the trench wall device is achieved in that the holding device is attached to a crossbeam which is arranged in an upper region of the base frame. In this way, the holding device, which can be a connection element or a coupling element for attaching a support cable or support bar, can be positioned centrally and/or along a longitudinal axis of the trench wall device. By means of the at least one crossbeam, a good connection to external adjacent longitudinal beams can be carried out in this central arrangement.

[0018] Another preferred embodiment of the invention resides in that the bracing system comprises at least two actuating elements arranged perpendicularly to the longitudinal direction, wherein at least one first actuating element is supported on the intermediate frame on the one side and, on the other side, is supported on a first bracing element and at least one second bracing element is supported on the intermediate frame on the one hand and on a second bracing element arranged opposite the first bracing element on the other hand, and that the first actuating elements and the second actuating elements are extendable perpendicular to the longitudinal direction.

[0019] At least one separate actuating element, in particular a bracing cylinder, can be assigned to each plate-shaped bracing element, and is supported on the relevant bracing element on the one side and, on the other side, is supported on the intermediate frame. The actuating elements are directed transversely to the longitudinal direction or propulsion direction of the trench wall device here. As a result, the individual bracing elements can be individually adjusted relative to the intermediate frame in a targeted manner. In addition, a clear separation is achieved between the transverse forces and the longitudinal forces for pressing-on the trench wall device, which overall contributes to a compact structure and efficient operation. In principle, opposite bracing elements can also be attached to a shared holder or an actuating device, wherein the holder is floatingly mounted on the base frame, so as to be movable in a transverse and/or longitudinal direction.

[0020] According to a continuing configuration of the invention, a compact structure is still promoted in that the intermediate frame is displaceably mounted on the base frame on a mounting portion, which has a reduced cross-section with respect to the base frame, while forming at least one free space, and in that at least one receptacle is formed on the displaceable intermediate frame for receiving at least one part of the actuating element, wherein the receptacle is arranged in the free space formed. There is therefore sufficient installation space for the arrangement of actuating elements, which are actuators or linear drives, in particular bracing cylinders, preferably hydraulic cylinders.

[0021] For a symmetrical application of force, it is also advantageous that two lateral free spaces are formed on the mounting portion and that two receptacles for actuating elements are accordingly formed on the intermediate frame. Therefore, the actuating elements for the opposite bracing elements are arranged in the two lateral receptacles. In this case, at least one first actuating element and at least one second actuating element for the respectively opposite bracing elements are provided in each lateral receptacle.

[0022] Preferably, the receptacle can be configured to be open and to just comprise a base plate, to which the relevant actuating element is fastened and on which it is supported. Opposite actuating elements can be mutually supported on the same base plate, wherein the transverse forces on the base plate are largely compensated for. According to another embodiment variant of the invention, it is advantageous that the at least one receptacle on the intermediate frame is of box-type design

and comprises at least one first actuating element and at least one second actuating element. In this case, the box-type receptacle comprises side walls, which face the respective plate-shaped bracing elements, wherein the side walls each comprise at least one passage for at least one of the actuating elements and at least one support region for attaching the actuating element directed towards the opposite bracing element.

[0023] For applying particularly high tensile forces to the bracing elements, according to a continuing configuration of the invention, it is preferable that multiple first actuating elements and multiple second actuating elements are arranged in one receptacle of the intermediate frame. In particular, two first actuating elements and two second actuating elements are arranged in each receptacle, which are each directed in opposite directions transversely to the longitudinal direction.

[0024] For targeted actuation of the individual actuating elements, according to a continuing configuration of the invention, it is advantageous that a controller is provided, by means of which the first actuating elements and the second actuating elements can be extended in a defined manner. In particular, the actuating elements can be uniformly extended, such that the individual bracing elements come to rest on the surrounding walls of the trench at the same time as far as possible, such that the position of the base frame of the trench wall device in the trench is not caused to change.

[0025] In particular, according to an embodiment variant of the invention, it is preferable that the controller is configured to first extend the at least one first actuating element and the at least one second actuating element with a first, low contact pressure force until the bracing elements abut the surrounding ground, and then to apply a high contact pressure force for the bracing. The high tensile force required for the bracing is, in particular only then initiated by the control device when no further extension movement is ascertained on the actuating elements at the first adjustment force. This can be taken as a sign that, during extending into a bracing position, the respective bracing elements have come into contact with the respectively adjacent wall.

[0026] In particular, it is advantageous that the control device is connected to sensors, by means of which a contact pressure force and/or an extension position of the actuating elements can be detected. Here, the contact pressure force can be detected at the individual actuating elements, or centrally for groups of actuating elements, or for all the actuating elements together.

[0027] Another preferred embodiment of the invention resides in that the pressing-on device is arranged above the bracing system and comprises at least one, preferably multiple pressing-on cylinders. In such an arrangement of the pressing-on device at the top, the bracing system can be arranged very close to the soil removal device. This allows the trench wall device to be braced in the trench formed at a very early stage, such that, even in an early phase of a foundation method, additional contact pressure forces can be applied to the soil removal device. Here, the contact pressing cylinder(s) is/are configured such that they exert a tensile force on the base frame to apply the contact pressure force. By contracting the at least one contact pressing cylinder, the base frame and therefore the soil removal device can thus be pulled downwards at a desired force.

[0028] A particularly stable embodiment can, according to a continuing configuration of the invention, be achieved in that each pressing-on cylinder is attached on the base frame on the one side and on the intermediate frame or on a bracing element on the other side. This allows for a direct flow of force during the pressing-on between the longitudinally displaceable base frame and the fixed intermediate frame or the relevant bracing element.

[0029] In principle, the at least one pressing-on cylinder can be arranged in parallel with the longitudinal direction of the trench wall device. According to an embodiment variant of the invention, a particularly compact arrangement results in that the pressing-on cylinders are arranged inclined to the vertical, wherein the base frame with the soil removal device can be pressed downwards relative to the braced intermediate frame by actuating the pressing-on cylinders.

[0030] A particularly expedient configuration of the invention resides in particular in that the trench wall device is configured as a trench cutter, wherein the soil removal device comprises at least one

cutting wheel. Preferably, two pairs of driven cutting wheels having cutting wheel axes that are parallel to one another are arranged at the lower end of the base frame as the soil removal device. [0031] An alternative configuration of the invention is a trench wall device that is configured as a trench wall grab, wherein the soil removal device comprises grab scoops. In a trench wall grab, a particularly high removal capacity can be achieved by the bracing system configured according to the invention, in particular when inserting and closing the grab scoops into the ground. [0032] The invention further comprises a method for creating a trench in the ground using the above-described trench wall device according to the invention, wherein the method is characterized in that the trench wall device is held on an upper end region by means of a holding device, in that soil is removed by means of at least one soil removal device, which is arranged on a lower end region of the base frame, while creating the trench, in that at least two opposite bracing elements are extended laterally by means of at least two bracing elements directed perpendicularly to the longitudinal direction by means of a bracing system, wherein at least part of the trench wall device is braced in the trench, in that the at least one soil removal device is applied with a contact-pressing force by means of a pressing-on device, in that the bracing system having the at least two bracing elements and the actuating elements is arranged on an intermediate frame, on which the holding device is attached, and the base frame is mounted displaceably with respect to the intermediate frame and the holding device, in that the intermediate frame is braced and secured with respect to the ground by means of the bracing system and in that the base frame with the soil removal device is pressed downwards and displaced relative to the intermediate frame and the holding device by means of the pressing-on device. [0033] By means of the method according to the invention, the above-described advantages can be achieved when operating a trench wall device according to the invention.

Description

[0034] The invention is described in greater detail in the following, on the basis of a preferred, exemplary embodiment, which is shown schematically in the drawings, in which show:

[0035] FIG. 1 a perspective view of a trench wall device according to the invention with retracted cutting wheels;

[0036] FIG. 2 a side view of the trench wall device according to FIG. 1;

[0037] FIG. 3 a front view of the trench wall device from FIGS. 1 and 2;

[0038] FIG. 4 a perspective view of the trench wall device according to the invention with extended cutting wheels;

[0039] FIG. 5 a perspective view of a detail of an upper region of the trench wall device according to the invention from FIG. 4;

[0040] FIG. 6 a perspective view of a detail of a lower region of the trench wall device according to the invention in the state according to FIG. 1 to 3;

[0041] FIG. 7 a perspective view of a detail of the lower region of the trench wall device according to the invention in the state according to FIG. 4; and

[0042] FIG. 8 a perspective view of a trench wall device according to the invention comprising a carrier unit.

[0043] A trench wall device **10** according to the invention comprising an elongate base frame **12** is shown in FIG. 1 to 3. The base frame **12**, which is preferably scaffold-like and rectangular in cross-section, can comprise a tapered mounting portion **16** at its lower end region which is likewise configured to have a basically rectangular cross-section. Relative to the cross-section of the base frame **12**, a long side can be shortened such that two lateral free spaces **17** are formed. The soil removal device **20** for removing soil material can be arranged on an underside of the mounting portion **16**.

[0044] In the present exemplary embodiment, by way of example the soil removal device **20** is formed by four cutting wheels **22**, two of which form a cutting wheel pair having a shared cutting wheel axis in each case. As is known per se, the cutting wheels **22** of a cutting wheel pair are each rotatably mounted and driven on a central bearing plate. As is also known per se, a suction box **18**, which is used to suction off cut-out soil material, can be arranged between the two pairs of cutting wheels **22**. The central suction line along the base frame **12** is not shown in greater detail for the sake of clarity.

[0045] An intermediate frame **60** with a bracing element **50** is longitudinally displaceably mounted on the mounting portion **16** of the base frame **12**. The intermediate frame **60** can comprise two front plates **62**, which can be interconnected by two outer side plates **64** and two inner side plates **66**, such that the box-type intermediate frame **60** surrounds the rectangular mounting portion **16** in the manner of a ring and is displaceably mounted thereon in the longitudinal direction.

[0046] An outer side plate **64** and an inner side plate can be arranged at a spacing from one another on each of the two short sides of the intermediate frame **60**, wherein a box-type receptacle **70** can be formed therebetween. Here, the two lateral receptacles **70** are located in the two free spaces **17**, which are formed by the tapering of the mounting portion **16** relative to the base frame **12**. Two first actuating elements directed towards the front and two second actuating elements directed towards the rear, which are, in particular configured as hydraulic cylinders, can each be arranged in the receptacles **70** for forming the bracing system **50**. By means of the actuating elements, lateral plate-shaped bracing elements **52** can be laterally extended and retracted in order to securely brace the trench wall device **10** in the cut trench.

[0047] By way of example, four pressing-on cylinders **32** are provided for forming a pressing-on device **30**. In this case, two pressing-on cylinders **32** are arranged on a front side and two pressing-on cylinders **32** are arranged on a rear side of the base frame **12**. Each pressing-on cylinder **32** is articulated to a bearing block **36**, which is fastened to the base frame **12**, by its cylinder housing. The piston rod of each pressing-on cylinder **32** is connected to the respectively associated plate-shaped bracing element **52** by means of a pivot pin connection **38**.

[0048] The pressing-on cylinders **32** can bring about displacement of the base frame **12** relative to the intermediate frame **60**. In this case, the cutting wheels **22** can be moved from an upwardly retracted position, which is shown in FIG. 1 to 3, into a lower position according to FIG. 4, in which the cutting wheels **22** project downwards relative to the intermediate frame **60** with the bracing system **50**. To do this, the pressing-on cylinders **32** can be retracted, as shown in FIG. 4, for example.

[0049] To hold the trench wall device **10**, a holding device **14** is arranged in an upper region, by means of which the trench wall device **10** can be fastened to a support cable or support bar. The holding device **14**, which is shown in greater detail in FIG. 5, can comprise a shaft **15** and, in this case, is mounted displaceably relative to the base frame **12** in the longitudinal direction by means of a sleeve-shaped linear guide **13** on the base frame **12** and is fastened to a crossbeam **42**. The crossbeam **42** is fixedly connected to the intermediate frame **60** with the bracing system **50** by means of one or multiple longitudinal beams **40**. Therefore, the trench wall device **10** is held by means of the holding device **14** and the intermediate frame **60**, wherein the base frame **12** is freely displaceable relative to the holding device **14** in the longitudinal direction by means of the pressing-on cylinders **32**. Guide plates **11** can be attached to the outer side of the base frame **12** for guidance within the cut trench. The guide plates **11** can be actuatable and displaceable, in order to orient the trench wall cutter **10** in the cut trench.

[0050] In the following, the mode of operation of the above-described trench wall device **10** and the method according to the invention for creating a trench in the ground will be explained in greater detail also in conjunction with FIGS. 6 and 7.

[0051] FIG. 6 shows a lower region of the trench wall device **10** according to the invention in a state corresponding to FIG. 1 to 3, in which the plate-shaped bracing elements **52** of the bracing

system **50** can be retracted such that the trench wall device **10** can be moved freely in a trench in the ground. The pressing-on cylinders **32** of the pressing-on device **30** are extended here, wherein the intermediate frame **60** is moved to a lower end of the mounting portion **16**.

[0052] The bracing system **50** is activated for bracing the trench wall device **10** in a trench in the ground. In this case, the actuating elements (which are not visible and are arranged in the receptacles **70**), preferably transversely directed hydraulic cylinders, are extended, wherein the plate-shaped bracing elements **52** are pressed against surrounding walls of soil.

[0053] Once the intermediate frame **60** is held on the surrounding walls of soil in a friction-locked manner by the bracing system **50**, the pressing-on device **30** can be activated, as shown in FIG. 7. Here, the pressing-on cylinders **32** are retracted, wherein, owing to the intermediate frame **60** being fixed in position by the extended bracing elements **52** by the pulling motion of the pressing-on cylinders **32** that is generated, the base frame **12** is pressed downwards together with the mounting portion **16** and the soil removal device **20**. In this way, in a suspended, height-adjustable trench wall device **10**, in addition to the weight force of the displaceably mounted base frame **12** and the soil removal device **20** an additional force can be applied to the soil removal device **20** in the downward direction by the pressing-on cylinders **32**. This can be advantageous when cutting hard ground, for example, in particular rock material.

[0054] The weight force of the trench wall device **10** can be absorbed by the bracing system **50** in full or in part and dissipated into the surrounding soil. This can be advantageous for soft soil, e. g. if the soil is highly compacted by an excessively high downward force, which can make removal difficult. The intermediate frame **60** with the bracing system **50** can be held on a support cable **5** of a carrier unit **1** by means of the longitudinal beams **40** and the holding device **14**, as shown in FIG. 8. The carrier unit **1** can be a mobile crane comprising a jib. The required liquids can be supplied and carried away by means of hose lines and supply lines and power can be supplied via the carrier unit **1**.

Claims

1. A diaphragm wall device for creating a trench in the ground, comprising a base frame which extends in a longitudinal direction, at least one soil removal device, which is arranged on a lower end region of the base frame, a holding device for holding the diaphragm wall device on its upper region, a bracing device with at least two opposite bracing elements displaceable laterally, transversely to the longitudinal direction and with at least one actuating element, by means of which the bracing elements can be extended laterally in the trench for bracing at least a part of the diaphragm wall device, an intermediate frame, on which the bracing device with the at least two bracing elements and the at least one actuating element is arranged, and a pressing-on device, by means of which the at least one soil removal device can be applied with a pressure force, wherein the pressing-on device is arranged between the intermediate frame and the base frame, so that the base frame is displaceable relative to the intermediate frame by means of the pressing-on device, wherein the holding device is fixedly connected with the intermediate frame, and wherein the base frame with the soil removal device is displaceable with respect to the holding device, wherein the intermediate frame is displaceably mounted in a lower region of the base frame along the base frame and the holding device is fixedly connected to the intermediate frame via one or multiple longitudinal beams, which extend along the base frame.

2. The diaphragm wall device according to claim 1, wherein the holding device is attached to a crossbeam which is arranged in an upper region of the base frame and a linear guide is arranged in the region of the crossbeam, by means of which the base frame is displaceably guided with respect to the holding device.

3. The diaphragm wall device according to claim 1, wherein the bracing device comprises at least two actuating elements arranged perpendicularly to the longitudinal direction, wherein at least one

first actuating element is mounted on the intermediate frame on the one side and, on the other side, is mounted on a first bracing element and at least one second bracing element is mounted on the intermediate frame on the one hand and is mounted on a second bracing element arranged opposite the first bracing element on the other hand, and the first actuating element and the second actuating element are extendable perpendicular to the longitudinal direction.

4. The diaphragm wall device according to claim 1, wherein the intermediate frame is displaceably mounted on the base frame in a lower region on a mounting portion, which has a reduced cross-section with respect to the base frame, while forming at least one free space.

5. The diaphragm wall device according to claim 4, wherein two lateral free spaces are formed on the mounting portion and two receptacles for actuating elements are accordingly formed on the intermediate frame.

6. The diaphragm wall device according to claim 5, wherein the at least one receptacle on the intermediate frame is of box-type design and comprises at least one first actuating element and at least one second actuating element.

7. The diaphragm wall device according to claim 1, wherein a controller is provided, by means of which the first actuating elements and the second actuating elements can be extended in a defined manner.

8. The diaphragm wall device according to claim 1, wherein the pressing-on device is arranged above the bracing device and comprises one, preferably multiple pressing-on cylinders.

9. The diaphragm wall device according to claim 8, wherein each pressing-on cylinder is attached on the base frame on the one side and on the intermediate frame or on a bracing element on the other side.

10. The diaphragm wall device according to claim 8, wherein the pressing-on cylinders are arranged inclined to the vertical, wherein the base frame with the soil-removal device can be pressed downwards relative to the braced intermediate frame by actuating pressing-on cylinders.

11. The diaphragm wall device according to claim 1, wherein the device is configured as a diaphragm wall cutter, wherein the soil removal device comprises at least one cutting wheel.

12. The diaphragm wall device according to claim 1, wherein the device is configured as a diaphragm wall grab, wherein the soil removal device comprises grabber blades.

13. A method for creating a trench in the ground using a diaphragm wall device according to claim 1, wherein the diaphragm wall device is held on an upper end region by means of a holding device, soil is removed by means of at least one soil removal device, which is arranged on a lower end region of the base frame, while creating a trench, by means of a bracing device at least two opposite bracing elements are extended laterally by means of at least two actuating elements oriented perpendicularly to the longitudinal direction, wherein at least a part of the diaphragm wall device is braced in the trench, the at least one soil removal device is applied with a pressing-on force by means of a pressing-on device, the bracing device having the at least two bracing elements and the actuating elements is arranged on an intermediate frame, on which the holding device is attached, and the base frame is mounted displaceably with respect to the intermediate frame and the holding device, the intermediate frame is braced and secured with respect to the ground by the bracing device and the base frame with the soil removal device is pressed downwards and is displaced relative to the intermediate frame and the holding device by means of the pressing-on device.
